Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

END SEMESTER EXAMINATION (2025-26 Even)

B. E. - IV Year (COE / CSE / CSBS), Semester VIII
May 14, 2026, Thursday, 09:00 – 12:00 Noon
Time: 3 Hours,
Name Of Faculty: Dr. Anju Bala, Dr. Nitin Saxena

Course Code: UCS813 / UCT802
Course Name: Social Network Analysis
M. Marks: 40

Note: *Attempt **any five** questions. Assume missing data, if any, suitably.*

Q1(a) In sparse graphs, the number of edges is often close to the number of nodes. Give an example of a connected graph where adding a single edge decreases the diameter by more than half. Use ceiling function for rounding off the values. (4)

(b) A bipartite graph has partitions X and Y where, |X|=7 and |Y|=9. If every vertex in X has degree 4. (4)

i. Determine whether such a graph is possible. Justify your answer.
ii. Find the total number of edges and average degree of vertices in Y.

Q2(a) A customer chooses between two coffee shops: A and B. If they visit A today, there is a 70% chance they return to A tomorrow. If they visit B today, there is a 40% chance they switch to A tomorrow. Find the long-run proportion of visits to each shop. (4)

(b) Consider an on-line news site, which consists of a front page with links to many different articles. The people who operate such sites generally track the popularity of the various articles that get posted, asking questions like the ones that we've seen: "As a function of k, what fraction of all articles have been viewed by k people?" Let's call this the popularity distribution of the articles. Now suppose that the operators of such a news site are considering changing the front page, so that next to each link is a counter showing how many people have clicked on the link. (E.g., next to each link it will say something like, "30,480 people have viewed this story," with the number getting updated over time.) (4)

What effect do you think this change will have on the behaviour of people using the site? Do you expect that adding this feature will cause the popularity distribution of the articles to follow a power-law distribution more closely or less closely compared to the version of the site before these counters were added? Give an explanation for your answer.

Q3(a) Apply Clique Percolation Method (CPM) on the following graph shown in *figure 1.* Use clique of size (k) =3 for identifying communities. (4)

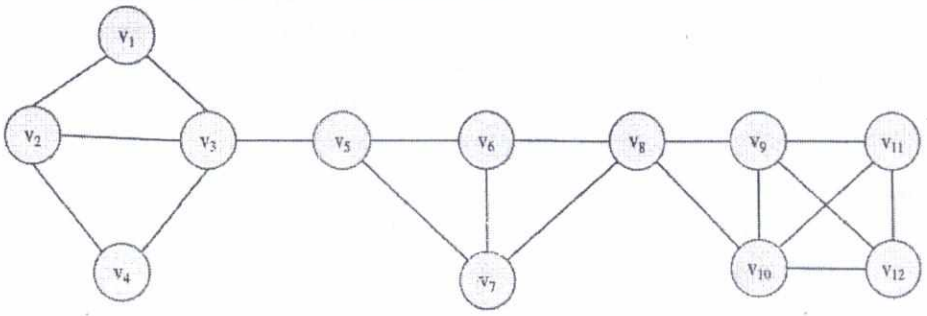


Figure 1

(b) Differentiate between: (4)

i. Modularity and Conductance
ii. Community Detection and Graph Partitioning

Q4 Consider a book club with 5 members and the ratings given by them to 4 books as shown in *Table 1*. It is observed that the rating given by User3 for Book2 is missing. Use user-based collaborative filtering method to predict the missing value based on the ratings given by other book club members who have similar reading preferences as User3. Use neighbourhood size =2 to identify similar user and normaliser constant = 1. (8)

	Book1	Book2	Book3	Book4
User1	5	4	0	2
User2	3	0	3	3
User3	3	?	1	0
User4	1	2	4	2
User5	2	2	0	1

Table 1

Q.5(a) Consider the social network represented in the *figure 2*. Suppose that this social network was obtained by observing a group of people at a particular point in time and recording all their friendship relations. Now suppose that we come back at some point in the future and observe it again. According to the theories based on empirical studies of triadic closure in networks, which new edge is most likely to be present? Also, give a brief explanation for your answer. (4)

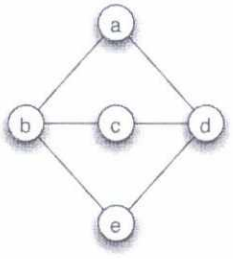


Figure 2

(b) Identify the primary event type involved in the following statements. Justify your answer. (4)

i. Bank updates account balance immediately after ATM withdrawal.
ii. Income tax department processes yearly returns after deadline.
iii. Passport application form submitted at service center.
iv. Weekly attendance report generated every Sunday.

Q6(a) The "*six degrees of separation*" concept states that it is likely most pairs of people in the world are connected by a path of at most six edges in the social network, where an edge joins any two people who know each other. Now consider a variation of this; For each person in the world, we ask them to rank the 30 people they know best, in descending order of how well they know them (It is assumed that each person is able to think of 30 people to list). We then construct a different social network, the "close-friend" network in which from each person we create a directed edge only to their ten closest friends on the list. In the close-friend network, is it likely that for each pair of people in the world, there is a path of at most six edges? Justify your answer. (4)

(b) A new employee in a company meeting agrees with senior staff about a technical issue because everyone else appears confident and experienced. Later, the employee discovers the decision was incorrect. (4)

i. Identify the type of social influence involved.
ii. Why was the influence effective?
iii. How can organizations reduce such conformity errors?

********* End of Paper *********